# Binomial Asset Pricing Model
Volume 1, Chapter 1

## §1. Binomial Asset Pricing Model

### §1.1. Notation

- **Time 0**: Very start of the period. **Time 1**: End of the first period.
- $S_0$: Stock price at time 0. $S_1(H)$ is price at time 1 if coin flip is heads, $S_1(H)$ is price at time 1 if coin flip i tails.
- $p$ is probability of heads, $q = 1 - p$ is probability of tails.
- $r$ is the interest rate, where $1$ dollar invested at time zero yields $1 + r$ at time one.
- $K$ is the strike price for European call options.

### §1.2. One-Period Binomial Model

**Main Idea**: Consider the following diagram.

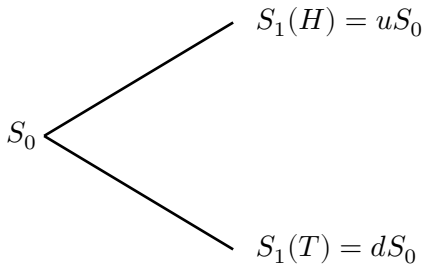



Figure 1.2.1: One-period binomial model

We know the outcome of the coin toss at time 1, but not time 0. Define

$$u = \frac{S_1(H)}{S_0}, \quad d = \frac{S_1(T)}{S_0}.$$

with $d < u$.

**Definition 1.2.1** (Arbitrage).
A trading strategy that begins with no money, has zero probability of losing money, and positive probability of making money.

**Remark 1.2.1.**
We assume that there is no arbitrage in our analysis. Otherwise, we get contradictions because we can basically get infinite wealth generation.

**Remark 1.2.2.**
To prevent arbitrage, we must have $0 < d < 1 + r < u$.

Otherwise, If $d \geq 1 + r$, borrow money and buy the stock with the borrowed money, and the stock will always earn enough to pay off the borrowed money. If $u \leq 1 + r$, short the stock and use the proceeds in the money market, which will earn enough interest to buy back the stock at time 1. The converse is true.

Stock prices obviously do not move like this, but given a sufficient number of periods, this helps approximate continuous-time models.

**Definition 1.2.2** (European Call Option)**.**
A European call option gives its owner the right, but not the obligation, to buy one share of the stock at time one for strike price $K$ at a certain point in time.

We consider the case where $S_1(T) < K < S_1(H)$. Observe that if we flip tails, the option expires worthless, but if we flip heads, the option has a payoff of $S_1(H) - K$.

**Definition 1.2.3** (Arbitrage Pricing Theory)**.**
Pricing options by replicating the option by trading in the stock and money markets.

**Remark 1.2.3.**
If we can get the same returns using the stock and money markets as we get from the option, then they should be valued at the same price.

**Example 1.2.1.**
Consider the following:

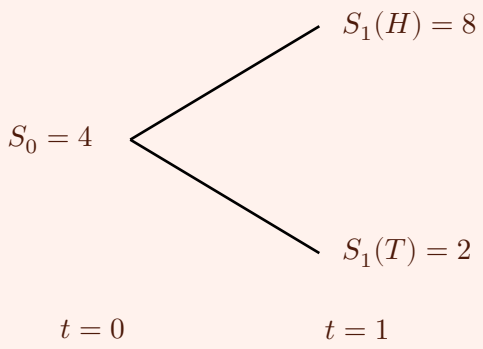



Figure 1.2.2: Example one-period binomial model

With $r = \frac{1}{4}$ and $K = 5$. Observe that if we have an initial wealth of $X_0 = 1.20$, borrow $0.80$ to spend $2.00$ on $\frac{1}{2}$ shares at time zero, then

$$X_1(H) = \frac{1}{2}S_1(H) - (1 + r)(0.80) = 3$$

and

$$X_1(T) = \frac{1}{2}S_1(T) - (1 + r)(0.80) = 0$$

which match the payoffs of the option! We have replicated the European call option, meaning that the option **must be worth the initial wealth** of $1.20$.

If the option was worth more than $1.20$, we could short the option and buy the replicating portfolio (i.e. buy $\frac{1}{2}$ shares and borrow $0.80$). If the option was worth less than $1.20$, we could buy the option and short the replicating portfolio (i.e. short sell $\frac{1}{2}$ shares and invest $0.80$ in the money market). Both cases result in an **arbitrage opportunity**.

There are a few assumptions made:
- We can buy fractional shares.
- Interest rate for investing is the interest rate for borrowing.
- Purchase price of stock is the same as its selling price.
- Stock prices can be only one of two values in the next period.

**Definition 1.2.4** (Derivative Security)**.**
A derivative security is a security that pays, at time 1, $V_1(H)$ if the coin toss is heads and $V_1(T)$ if the coin toss is tails

To get the price of a derivative security at time 0, we replicate the security. We can derive a formula for this. Let $X_0$ be the value of the replicating portfolio, and hence, the **value of the option**. Let $\Delta_0$ be the number of shares of stock we buy at time 0. The value of the replicating portfolio at time 1 is

$$X_1 = \underbrace{\Delta_0 S_1}_{\text{Position in stocks}} + \underbrace{(1+r)(X_0 - \Delta_0 S_0)}_{\text{Amount due on borrowed}} = \Delta_0(S_1 - (1+r)S_0) + (1+r)X_0$$

We want to find $X_0$ and $\Delta_0$ so that $X_1(T) = V_1(T)$ and $X_2(H) = V_2(H)$.

Define $\tilde{p}$ and $\tilde{q} = 1 - \tilde{p}$ to be the values satisfying

$$S_0 = \frac{1}{1+r}[\tilde{p}S_1(H) + \tilde{q}S_1(T)]$$

This is similar to calculating the expected value for $S_0$. Equating $X_1(H) = V_1(H)$ and $X_1(T) = V_1(T)$, we have

$$\begin{aligned} X_0 + \Delta_0\left(\frac{1}{1+r}S_1(H) - S_0\right) &= \frac{1}{1+r}V_1(H) \\ X_0 + \Delta_0\left(\frac{1}{1+r}S_1(T) - S_0\right) &= \frac{1}{1+r}V_1(T) \end{aligned} \tag{$*$}$$

We can add these equations together, multiplied by $\tilde{p}$ and $\tilde{q}$, to get

$$X_0 + \Delta_0\left(\frac{1}{1+r}[\tilde{p}S_1(H) + \tilde{q}S_1(T)] - S_0\right) = \frac{1}{1+r}[\tilde{p}V_1(H) + \tilde{q}V_1(T)].$$

Because of how we defined $\tilde{p}$ and $\tilde{q}$, we get

$$X_0 = X_0 + \Delta_0(S_0 - S_0) = \frac{1}{1+r}[\tilde{p}V_1(H) + \tilde{q}V_1(T)].$$

To solve for $\tilde{p}$, we have

$$S_0 = \frac{1}{1+r}(\tilde{p}uS_0 + (1-\tilde{p}dS_0)$$
$$1 + r = \tilde{p}u + (1-\tilde{p})d$$

Which gives us

$$\tilde{p} = \frac{1+r-d}{u-d}.$$

Hence, we obtain

$$\tilde{q} = \frac{u-1-r}{u-d}.$$

To solve for $\Delta_0$, we subtract the equalities in $*$ to obtain

$$\Delta_0 = \frac{v_1(H) - V_1(T)}{S_1(H) - S_1(T)}$$

Therefore, we should value the security at

$$V_0 = \frac{1}{1+r}[\tilde{p}V_1(H) + \tilde{q}V_1(T)]$$

**Definition 1.2.5** (Risk Neutral Probabilities)**.**
We call $\tilde{p}$ and $\tilde{q}$ the risk-neutral probabilities.

**Remark 1.2.4.**
$\tilde{p}$ and $\tilde{q}$ are indeed probabilities as they sum to $1$ and are non-negative. They represent the probabilities of the coin flip so that the **expected** growth rate of the stock is equal to $r$. In fact, because investors compensate for risk, if $p$ and $q$ represent the actual probabilities of the coin toss,

$$S_0 < \frac{1}{1+r}[pS_1(H) + qS_1(T)]$$
$$\frac{1}{1+r}[\tilde{p}S_1(H) + \tilde{q}S_1(T)] < \frac{1}{1+r}[pS_1(H) + qS_1(T)]$$

**Remark 1.2.5.**
We do not worry about the actual probabilities for valuation! We have constructed a replicating portfolio for the option, so the probabilities of up and down are irrelevant. Our replicating portfolio must earn the risk-free return.

## §1.3. Multiperiod Binomial Model

What happens if we have multiple periods?

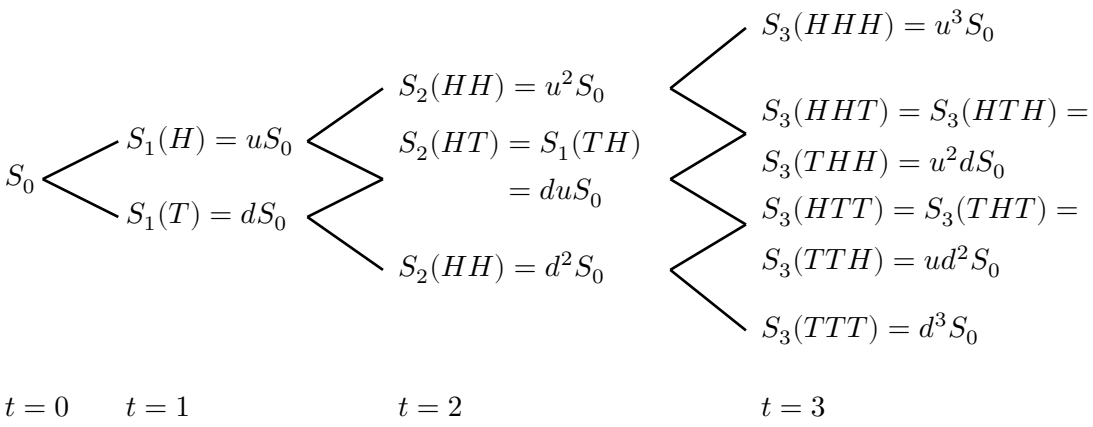



Figure 1.3.3: Three-period binomial model

To get the price using an $N$-period binomial model, we can work backwards. The idea is that at each time, we readjust the replicating portfolio/hedge to get risk-free returns. We can therefore extend the one-period case recursively.

More precisely, the wealth equation gives us the value of the replicating portfolio starting at time $t = 0$:

$$X_{n+1} = \underbrace{\Delta_n S_{n+1}}_{\text{Value of shares held since } t=n \text{ at new price}} + \underbrace{(1+r)(X_n - \Delta_n S_n)}_{\text{Money market investment}}.$$

To replicate a derivative with value $V_n$, we want to find the corresponding $\Delta_i$ (for all nodes of the tree) and value of $X_i$ so that

$$X_n(\omega_1 ... \omega_n) = V_n(\omega_1 ... \omega_n)$$

for any outcomes $\omega_i$. In other words, we look for a hedging strategy so that in the case of any outcome of up and down movements (the coin flips), **the value of the replicating portfolio is equal to the value of the derivative**. We get the following theorem by applying the one-period model:

**Theorem 1.3.1** (Replication in the multiperiod binomial model)**.**
Consider an $N$-period binomial asset-pricing model, with $0 < d < 1 + r < u$ (to prevent arbitrage), and define

$$\tilde{p} = \frac{1+r-d}{u-d}, \quad \tilde{q} = \frac{u-1-r}{u-d}.$$

Let $V_N$ be a random variable (representing the value of a derivative security's payoff at time $N$) which is dependent on the first $N$ coin tosses $\omega_1\omega_2...\omega_n$. For $i = 0, 1, ..., N-1$, define

$$V_i(\omega_1 ... \omega_i) = \frac{1}{1+r}\left[\tilde{p}V_{n+1}(\omega_1 ... \omega_i H) + \tilde{q}V_{n+1}(\omega_1 ... \omega_i T)\right]$$

and also set

$$\Delta_i(\omega_1 ... \omega_i) = \frac{V_{n+1}(\omega_1 ... \omega_i H) - V_{n+1}(\omega_1 ... \omega_i T)}{S_{n+1}(\omega_1 ... \omega_i H) - S_{n+1}(\omega_1 ... \omega_i T)}.$$

Forward recursively define

$$X_{i+1} = \Delta_i S_{i+1} + (1+r)(V_i - \Delta_i S_i)$$

Then,

$$X_N(\omega_1...\omega_N) = V_N(\omega_1...\omega_N)$$

for all $\omega_1...\omega_N$.

**Definition 1.3.1** (Price of Derivative)**.**
Given that the first $n$ tosses are $\omega_1...\omega_n$, define the price of the derivative security at time $n$ to be $V_n(\omega_1...\omega_n)$. Therefore, the price of the derivative security at time zero is $V_0$.

**Example 1.3.1.**
Let $u = 2$, $d = \frac{1}{2}$, $N = 3$, $S_0 = 4$, and $r = \frac{1}{4}$. Then, $\tilde{p} = \tilde{q} = \frac{1}{2}$. Consider a lookback option that pays $V_3 = \max_{0\le n\le 3} S_n - S_3$. Then,

$$\begin{aligned}
V_3(HHH) &= S_3(HHH) - S_3(HHH) &&= 32 - 32 = 0\\
V_3(HHT) &= S_3(HH) - S_3(HHT) &&= 16 - 8 = 8\\
V_3(HTH) &= S_3(H) - S_3(HHT) &&= 8 - 8 = 0\\
V_3(HTT) &= S_3(H) - S_3(HTH) &&= 8 - 2 = 6\\
V_3(THH) &= S_3(THH) - S_3(HTT) &&= 8 - 8 = 0\\
V_3(THT) &= S_3(TH) - S_3(THT) &&= 4 - 2 = 2\\
V_3(TTH) &= S_0 - S_3(TTH) &&= 4 - 2 = 2\\
V_3(TTT) &= S_0 - S_3(HHT) &&= 4 - 0.50 = 3.50
\end{aligned}$$

We get

$$\begin{aligned}
V_2(HH) &= \frac{4}{5}\left[\frac{1}{2}V_3(HHH) + \frac{1}{2}V_3(HHT)\right] = 3.20\\
V_2(HT) &= \frac{4}{5}\left[\frac{1}{2}V_3(HTH) + \frac{1}{2}V_3(HTT)\right] = 2.40\\
V_2(TH) &= \frac{4}{5}\left[\frac{1}{2}V_3(THH) + \frac{1}{2}V_3(THT)\right] = 0.80\\
V_2(TT) &= \frac{4}{5}\left[\frac{1}{2}V_3(TTH) + \frac{1}{2}V_3(TTT)\right] = 2.20
\end{aligned}$$

and

$$\begin{aligned}
V_1(H) &= \frac{4}{5}\left[\frac{1}{2}V_3(HH) + \frac{1}{2}V_3(HT)\right] = 2.24\\
V_1(T) &= \frac{4}{5}\left[\frac{1}{2}V_3(TH) + \frac{1}{2}V_3(TT)\right] = 1.20
\end{aligned}$$

giving us an option value of

$$V_0 = \frac{4}{5}\left[\frac{1}{2}V_1(H) + \frac{1}{2}V_1(T)\right] = 1.376.$$

**Proof of Theorem**: We prove by induction on $n$ that $X_n(\omega_1...\omega_n) = V_n(\omega_1...\omega_n)$. The base case, $n = 0$, is true because we defined $X_0 = V_0$.

For the induction step, assume that our hypothesis holds true for some $n < N$. We show that it holds for $n+1$. Fix a sequence of coin flips $\Omega = \omega_1 ... \omega_{n+1}$. If $\omega_{n+1} = H$, then

$$X_{n+1}(\Omega H) = (1+r)X_{n(\Omega)} - \Delta_n(\Omega)S_n(\Omega)(u-(1+r)). \qquad (*)$$

Our formula for $\Delta_n$ gives

$$\Delta_n = \frac{V_{n+1}(\Omega H) - V_{n+1}(\Omega T)}{S_{n+1}(\Omega H) - S_{n+1}(\Omega T)} = \frac{V_{n+1}(\Omega H) - V_{n+1}(\Omega T)}{uS_n(\Omega) - dS_n(\Omega)}$$

Substituting this into $(*)$, we get

$$\begin{aligned}
X_{n+1}(\Omega H) &= (1+r)X_n(\Omega) + \Delta_n S_n(\Omega)(u-(1+r)) \\
&= (1+r)V_n(\Omega) + \Delta_n S_n(\Omega)(u-(1+r)) && \text{Induction Hypothesis} \\
&= (1+r)V_n(\Omega) + \frac{V_{n+1}(\Omega H) - V_{n+1}(\Omega T)}{(u-d)S_n(\Omega)} S_n(\Omega)(u-(1+r)) && \text{Substitution} \\
&= (1+r)V_n(\Omega) + (V_{n+1}(\Omega H) - V_{n+1}(\Omega T))(\tilde{q}) && \tilde{q} = \frac{u-1-r}{u-d} \\
&= \tilde{p}V_{n+1}(\Omega H) + \tilde{q}V_{n+1}(\Omega T) + (V_{n+1}(\Omega H) - V_{n+1}(\Omega T))(\tilde{q}) && \text{Recursion for } V_n \\
&= \tilde{p}V_{n+1}(\Omega H) + \tilde{q}V_{n+1}(\Omega H) = V_{n+1}(\Omega H)
\end{aligned}$$

as required. □

## §1.4. Computational Considerations

- If we compute naively for $V_0$, where we do the full recursion for each node, the computational complexity is $\in O(2^N)$.
- Observe that at time $n$, there are only $n+1$ possible stock prices. Therefore, it is reasonable to store intermediate results, reducing computational complexity.
- The idea is that for each time, starting from the back, we compute the valuation given a price, i.e. we find $v_{n(s)}$ for time $n$ and stock price $s$, which should yield

$$v_{n(s)} = \frac{1}{1+r}[\tilde{p}v_{n+1}(us) + \tilde{q}v_{n+1}(ds)].$$

# Probability Theory (Finite)
Volume 1, Chapter 2

## §2. Probability Theory on Coin Toss Space

### §2.1. Finite Probability Spaces

We use **finite probability spaces** to model situations in a random experiment that permit **finitely many possible outcomes**.

**Example 2.1.1.**
If we toss a coin three times, the possble outcomes are

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

If the probability of a head is $p$ (and thus the probability of a tail is $1-p$), and we assume independent tosses, we get

$$\mathbb{P}(HHH) = p^3, \mathbb{P}(HHT) = \mathbb{P}(HTH) = \mathbb{P}(THH) = qp^2,$$
$$\mathbb{P}(HTT) = \mathbb{P}(THT) = \mathbb{P}(TTH) = q^2p, \mathbb{P}(TTT) = q^3$$

**Events** are subsets of $\Omega$.

**Example 2.1.2.**
The event "The first toss is a head" is the subset

$$\{\omega \in \Omega; \omega_1 = H\} = \{HHH, HHT, HTH, HTT\}$$

The probability of this event is

$$\begin{aligned}\mathbb{P}(\{\omega \in \Omega; \omega_1 = H\}) &= \mathbb{P}(HHH) + \mathbb{P}(HHT) + \mathbb{P}(HTH) + \mathbb{P}(THH) \\ &= p^3 + 2p^2q + pq^2 \\ &= p^2(p+q) + pq(p+q) \\ &= p^2 + pq \\ &= p\end{aligned}$$

More formally,

**Definition 2.1.1** (Finite Probability Space, Event)**.**
A finite probability space consists of a sample space $\Omega$, which is a non-empty finite set, and a probability measure $\mathbb{P} : \Omega \to [0,1]$ such that $\sum_{\omega\in\Omega} \mathbb{P}(\omega) = 1$.

An event is a subset of $\Omega$. The probability of an event $A$ is defined as $\mathbb{P}(A) = \sum_{\omega\in A} P(\omega)$

**Remark 2.1.1.**
Observe that

- $\mathbb{P}(\Omega) = 1$.
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ if $A \cap B = \varnothing$.

## §2.2. Random Variables

**Definition 2.2.1** (Random Variables)**.**
Let $(\Omega, \mathbb{P})$ be a finite probability space. A random variable is a function $\Omega \to \mathbb{R}$. We sometimes permit random variables to take the values $+\infty$ or $-\infty$.

We sometimes use $\omega$ to denote the argument of all the random variables, even if $\omega$ is a sequence.

**Example 2.2.1.**
Toss a coin three times. Define the random variables $X = \text{Total number of heads}$ and $Y = \text{Total number of tails}$. Then,

$$\begin{aligned} X(HHH) &= 3 \\ X(HHT) = X(THH) = X(THH) &= 2 \\ X(HTT) = X(THT) = X(TTH) &= 1 \\ X(TTT) &= 0 \end{aligned}$$

and

$$\begin{aligned} Y(HHH) &= 0 \\ Y(HHT) = Y(THH) = Y(THH) &= 1 \\ Y(HTT) = Y(THT) = Y(TTH) &= 2 \\ Y(TTT) &= 3. \end{aligned}$$

If each sequence of coin tosses has probability $\frac{1}{8}$ of occuring, then

$$\begin{aligned} \mathbb{P}(X = 0) = P(TTT) &= \frac{1}{8} \\ \mathbb{P}(X = 1) = P(TTH) + P(THT) + P(HTT) &= \frac{3}{8} \\ \mathbb{P}(X = 2) = P(THH) + P(HTH) + P(HHT) &= \frac{3}{8} \\ \mathbb{P}(X = 3) = P(HHH) &= \frac{1}{8} \end{aligned}$$

We write $X = j$ as shorthand for $\{\omega \in \Omega : X(\omega) = j\}$. We refer to the table of probabilities that $X$ takes various values as the **distribution** of $X$. Distributions and random variables are different!

**Definition 2.2.2** (Expected Values)**.**
Let $X$ be a random variable defined on finite probabiltiy space $(\Omega, \mathbb{P})$. The expectation/expected value of $X$ is

$$\mathbb{E}X = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\omega)$$

When we use risk-neutral probabilities, we use the following notation:

$$\tilde{\mathbb{E}}X = \sum_{\omega \in \Omega} X(\omega)\tilde{\mathbb{P}}(\omega)$$

**Definition 2.2.3** (Variance)**.**
The variance of $X$ is $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$

**Remark 2.2.1.**
Expectation is linear, i.e. $\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y$

**Theorem 2.2.1** (Jensen's Inequality)**.**
Let $X$ be a random variable on a finite probability space and let $\varphi(x)$ bea convex function of a dummy variable $x$. Then

$$\mathbb{E}[\varphi(X)] \geq \varphi(\mathbb{E}X).$$

**Proof**: We first argue that

$$\varphi(x) = \max\{\ell(x); \ell \text{ is linear and } \ell(y) \leq \varphi(y)) \text{ for all } y \in \mathbb{R}\}$$

I.e. $\varphi(x)$ is the maximum value attained at $x$ of all linear functions that lie below it. It is clear that $\varphi(x) \geq \max(...)$ because we only consider linear functions that lie below $\varphi$. To see that $\varphi(x) \leq \max(...)$, fix $x$ and consider the linear function $\ell$ that lies below $\varphi$ but $\ell(x) = \varphi(x)$, which exists because $\varphi$ is convex.

Let $\ell$ be a linear function below $\varphi$. Then,

$$\mathbb{E}[\varphi(X)] \geq \mathbb{E}[\ell(X)] = \ell[EX]$$

by linearity of expectation. Because this is true for all $\ell$, we get

$$\mathbb{E}[\varphi(X)] \geq \max\{\ell(\mathbb{E}X); \ell \text{ is linear and } \ell(y) \leq \varphi(y) \text{ for all } y \in \mathbb{R}\} = \varphi(\mathbb{E}x)$$

as required. □

**Example 2.2.2.**
Because $\varphi(x) = x^2$ is convex, $E[X^2] \geq \mathbb{E}[X]^2$.

## §2.3. Conditional Expectations

**Definition 2.3.1** (Conditional Expectation based on Information at Time $n$)**.**
Let $n$ satisfy $1 \leq n \leq N$ and let $\omega_1...\omega_n$ be given and fixed. Let $\#H(\omega_{n+1}...\omega_N)$ be the number of heads in $\omega_{n+1}...\omega_N$ and let $\#T(\omega_{n+1}...\omega_N)$ be the number of tails in $\omega_{n+1}...\omega_N$. Define

$$\tilde{\mathbb{E}}_{n[X]}(\omega_1 ... \omega_n) = \sum_{\omega_{n+1} ... \omega_N} \tilde{p}^{\#H(\omega_{n+1} ... \omega_N)} \tilde{q}^{\#T(\omega_{n+1} ... \omega_N)} X(\omega_1 ... \omega_N).$$

We call $\tilde{\mathbb{E}}_{n[X]}$ the conditional expectation of $X$ based on the information at time $n$.

For the extreme cases, define $\tilde{\mathbb{E}}_0[X] = \tilde{\mathbb{E}}X$ and $\tilde{\mathbb{E}}_N[X] = X$.

**Remark 2.3.1.**
The conditional expectation is basically the value at a single node of our binomial tree. For example, if $\omega_1 = H$, then $\tilde{E}_1[X](\omega_1)$ is the value assuming that we took an up action in our tree at the very start.

**Theorem 2.3.1** (Fundamental properties of conditional expectations)**.**
Let $N \in \mathbb{Z}_+$. Let $X$ and $Y$ be random variables dependent on the first $N$ coin tosses. Let $0 \leq n \leq N$ be given. The following properties hold.

1. **Linearity of conditional expectations**. For all constants $c_1$ and $c_2$,
$$\mathbb{E}_n[c_1 X + c_2 Y] = c_1 \mathbb{E}_n[X] + c_2 \mathbb{E}_n[Y].$$
2. **Taking out what is known**. If $X$ only depends on the first $n$ coin tosses,
$$\mathbb{E}_n[XY] = X\mathbb{E}_n[Y]$$
3. **Iterated conditioning**. If $0 \leq n \leq m \leq N$, then
$$\mathbb{E}_n[\mathbb{E}_m[X]] = \mathbb{E}_n[X].$$
This also implies that $\mathbb{E}[\mathbb{E}_m[X]] = \mathbb{E}X$
4. **Independence**. If $X$ only depends on coin tosses $n+1$ to $N$, then
$$\mathbb{E}_n[X] = \mathbb{E}X$$
5. **Conditional Jensen's inequality**. If $\varphi(x)$ is convex, then
$$\mathbb{E}_n[\varphi(X)] \geq \varphi(\mathbb{E}_n[X]).$$

## §2.4. Martingales

We have so far used risk-neutral probabilities so that for every coin toss sequence, we have that $S_n(\omega_1 ... \omega_n) = \frac{1}{1+r}[\tilde{p}S_{n+1}(\omega_1 ... \omega_n H) + \tilde{q}S_{n+1}(\omega_1 ... \omega_n T)] = \frac{1}{1+r}\tilde{\mathbb{E}}_n[S_{n+1}]$. We can divide by $(1+r)^n$ to get

$$\frac{S_n}{(1+r)^n} = \tilde{\mathbb{E}}_n\left[\frac{S_{n+1}}{(1+r)^{n+1}}\right] = \frac{1}{(1+r)^n}\tilde{\mathbb{E}}_n[S_{n+1}]$$

under the assumption that interest rates are non-random. This equation shows that the optimal forecast of the discounted stock price at time $n+1$ is the discounted stock price at time $n$. This quirk motivates the definition of a martingale

**Definition 2.4.1** (Martingale).
Consider the binomial asset-pricing model. Let $M_0, M_1, ..., M_N$ be a sequence of random variables, with each $M_n$ only depending on the first $n$ coin tosses (observe that $M_0$ is a constant). Such a sequence of random variables is called an **adapted stochastic process**.
1. If

$$M_n = \mathbb{E}_n[M_{n+1}]$$

for $n = 0, ..., N-1$, this process is a **martingale**.
2. If

$$M_n \leq \mathbb{E}_n[M_{n+1}]$$

for $n = 0, ..., N-1$, this process is a **submartingale**.
3. 2. If

$$M_n \geq \mathbb{E}_n[M_{n+1}]$$

for $n = 0, ..., N-1$, this process is a **supermartingale**.

**Remark 2.4.1.**
If $M_0, ..., M_N$ is a martingale and $n \leq N-2$, then we have

$$M_{n+1} = \mathbb{E}_{n+1}[M_{n+2}]$$

Using the iterated conditioning probability, $\mathbb{E}_n[M_{n+1}] = \mathbb{E}_n[\mathbb{E}_{n+1}[M_{n+2}]] = \mathbb{E}_n[M_{n+2}]$. We can iterate this to show that

$$M_n = \mathbb{E}_n[M_m].$$

In fact, because $\mathbb{E}M_n = \mathbb{E}[\mathbb{E}_n[M_{n+1}]] = \mathbb{E}[M_{n+1}]$, we get

$$\mathbb{E}M_0 = ... = \mathbb{E}M_N = M_0.$$

The main idea of a martingale is that given a sequence of events, the value of the state is equal to the expected value of the next state. Thus, the value of the martingale has **no tendency to rise or fall**. We find that stock prices with their actual probabilities for future prices are **submartingales**, i.e. their expected discounted prices at a future time are higher than their current prices, which makes sense given that investors usually enforce a risk premium.

**Theorem 2.4.1** (Risk-Neutral Probabilities create Martingales).
Let

$$\tilde{p} = \frac{1+r-d}{u-d}, \qquad \tilde{q} = \frac{u-1-r}{u-d}$$

with $0 < d < 1 + r < u$. Under these risk-neutral probabilities, the discounted stock price is a martingale.

**Proof**: Make the observation that $\frac{S_{n+1}}{S_n}$ depends only on the (n+1)st coin toss (it is either equal to $u$ or $d$ depending on the toss). With conditional probability properties, we get that

$$\begin{aligned}
\tilde{\mathbb{E}}_n\left[\frac{S_{n+1}}{(1+r)^{n+1}}\right] &= \tilde{\mathbb{E}}_n\left[\frac{S_n}{(1+r)^{n+1}} \cdot \frac{S_{n+1}}{S_n}\right] \\
&= \frac{S_n}{(1+r)^{n+1}}\tilde{\mathbb{E}}_n\left[\frac{S_{n+1}}{S_n}\right] && \text{Taking Out What is Known} \\
&= \frac{S_n}{(1+r)^{n+1}}\tilde{\mathbb{E}}\left[\frac{S_{n+1}}{S_n}\right] && \text{Independence} \\
&= \frac{S_n}{(1+r)^{n+1}}(\tilde{p}u + \tilde{q}d) \\
&= \frac{S_n}{(1+r)^n} && \square
\end{aligned}$$

Recalling our replicating portfolio idea, we see that $\Delta_0, ..., \Delta_{N-1}$ and $X_0, ..., X_{N-1}$ are both adapted stochastic process! In fact, there is something deeper about the wealth process:

**Theorem 2.4.2** (Discounted Wealth Process is a Martingale)**.**
Consider the binomal model with $N$ periods. Let $\Delta_0, ..., \Delta_{N-1}$ b an adapted portfolio process. Let $X_0 \in \mathbb{R}$ and let $X_1, ..., X_N$ be generated by

$$X_{n+1} = \Delta_n S_{n+1} + (1+r)(X_n - \Delta_n S_n).$$

Then $\frac{X_n}{(1+r)^n}$ for $n = 0, ...N$ is a martingale under the risk-neutral measure.

**Proof**: Observe

$$\begin{aligned}
\tilde{\mathbb{E}}_n\left[\frac{X_{n+1}}{(1+r)^{n+1}}\right] &= \tilde{\mathbb{E}}_n\left[\frac{\Delta_n S_{n+1}}{(1+r)^{n+1}} + \frac{X_n - \Delta_n S_n}{(1+r)^n}\right] \\
&= \tilde{\mathbb{E}}_n\left[\frac{\Delta_n S_{n+1}}{(1+r)^{n+1}}\right] + \tilde{\mathbb{E}}_n\left[\frac{X_n - \Delta_n S_n}{(1+r)^n}\right] && \text{Linearity} \\
&= \Delta_n\tilde{\mathbb{E}}_n\left[\frac{S_{n+1}}{(1+r)^{n+1}}\right] + \frac{X_n - \Delta_n S_n}{(1+r)^n} && \text{Taking out what is known} \\
&= \frac{\Delta_n S_n}{(1+r)^n} + \frac{X_n - \Delta_n S_n}{(1+r)^n} && \text{Theorem 2.4.1} \\
&= \frac{X_n}{(1+r)^n} && \square
\end{aligned}$$

**Corollary 2.4.1.**

$$\tilde{\mathbb{E}}\left[\frac{X_n}{(1+r)^n}\right] = X_0$$

**Proof**: Follows from Remark 2.4.1 as $\frac{X_n}{(1+r)^n}$ is a martingale.

**Remark 2.4.2.**
Our corollary shows that there is no arbitrage in the binomial model! Otherwise, suppose we

have a portfolio with $X_0 = 0$. If there was an arbitrage opportunity, $\tilde{\mathbb{E}}\left[\frac{X_n}{(1+r)^n}\right] > 0 = X_0$, a contradiction with our corollary.

**Remark 2.4.3.**
These ideas lead to the **First Fundamental Theorem of Asset Pricing**, which states that if there exists a risk-neutral measure in a model, i.e. a measure that agrees with the actual probabilities about the paths that have zero probability and where the risk-neutral discounted prices are martingales, then there is **no arbitrage.**

**Theorem 2.4.3** (Risk-neutral pricing formula)**.**
Consider $N$-period binomial asset-pricing model with $0 < d < 1 + r < u$ and risk-neutral probability measure $\tilde{\mathbb{P}}$. Let $V_n$ be a random variable (the value of a derivative security with payoff at time $N$) dependent on the coin tosses. The price of the derivative security is then given by

$$V_n = \tilde{\mathbb{E}}_n\left[\frac{V_N}{(1+r)^{N-n}}\right]$$

and the discounted price of the derivative security is a martingale under $\tilde{\mathbb{P}}$, i.e.

$$\frac{V_n}{(1+r)^n} = \tilde{\mathbb{E}}_n\left[\frac{V_{n+1}}{(1+r)^{n+1}}\right], n = 0, ..., N.$$

We define $V_n$ here identically as in Theorem 1.3.1.

**Proof**: Exercise 8

**Theorem 2.4.4** (Cash flow valuation)**.**
Consider $N$-period binomial asset-pricing model with $0 < d < 1 + r < u$ and risk-neutral probability measure $\tilde{\mathbb{P}}$. Let $C_0, ..., C_N$ be a sequence of random variables such that $C_n$ depends only on $\omega_1 ... \omega_n$. The price of a security at time $n$ that pays $C_n, ..., C_N$ at times $n, ..., N$ respectively is

$$V_n = \tilde{\mathbb{E}}_n\left[\sum_{k=n}^{N} \frac{C_k}{(1+r)^{k-n}}\right], \quad n = 0, ..., N.$$

The price process $V_n, n = 0, ..., N$ satisfies

$$C_{n(\omega_1 ... \omega_n)} = V_{n(\omega_1 ... \omega_n)} - \frac{1}{1+r}\left[\tilde{p}V_{n+1}(\omega_1 ... \omega_n H) + \tilde{q}V_{n+1}(\omega_1 ... \omega_n T)\right]$$

Define

$$\Delta_n(\omega_1 ... \omega_n) = \frac{V_{n+1}(\omega_1 ... \omega_n H) - V_{n+1}(\omega_1 ... \omega_n T)}{S_{n+1}(\omega_1 ... \omega_n H) - S_{n+1}(\omega_1 ... \omega_n T)}$$

for $n = 0, ...N - 1$. If we set $X_0 = V_0$ and recursively define $X_1, ..., X_N$ by

$$X_{n+1} = \Delta_n S_{n+1} + (1+r)(X_n - C_n - \Delta_n S_n)$$

we get

$$X_n(\omega_1...\omega_n) = V_n(\omega_1...\omega_n)$$

for all $n$ and $\omega_1...\omega_n$

**Remark 2.4.4.**
To finance practitioners, $V_n$ is known as the **net present value** (NPV) of the sequence of payments $C_n, ..., C_N$. Note that we include the payment at time $n$.

**Proof**: By induction on $n$. The base case is true by definition of $X_0$. Suppose that $X_{n(\omega_1...\omega_n)} = V_{n(\omega_1...\omega_n)}$ for some $n = 0, ..., N-1$. We show that $X_{n+1}(\omega_1...\omega_n H) = V_{n+1}(\omega_1...\omega_n H)$ and a similar argument demonstrates that $X_{n+1}(\omega_1...\omega_n T) = V_{n+1}(\omega_1...\omega_n T)$, which in turn proves our theorem.

Because $C_n$ depends on the first $n$ coin tosses only, we can modify the equation for $V_n$:

$$\begin{aligned}
V_n &= \tilde{\mathbb{E}}_n\left[\sum_{k=n}^{N}\frac{C_k}{(1+r)^{k-n}}\right] \\
&= C_n + \tilde{\mathbb{E}}_n\left[\frac{1}{1+r}\sum_{k=n+1}^{N}\frac{C_k}{(1+r)^{k-(n+1)}}\right] \\
&= C_n + \tilde{\mathbb{E}}_n\left[\frac{1}{1+r}\tilde{\mathbb{E}}_{n+1}\left[\sum_{k=n+1}^{N}\frac{C_k}{(1+r)^{k-(n+1)}}\right]\right] \\
&= C_n + \tilde{\mathbb{E}}_n\left[\frac{1}{1+r}V_{n+1}\right]
\end{aligned}$$

This leads to the following:

$$\begin{aligned}
V_n - C_n &= \tilde{\mathbb{E}}_n\left[\frac{1}{1+r}V_{n+1}\right] \\
&= \frac{\tilde{p}V_{n+1}(\omega_1...\omega_n H) + \tilde{q}V_{n+1}(\omega_1...\omega_n T)}{1+r}
\end{aligned}$$

For the rest of the proof, we suppress $\omega_1...\omega_n$ for ease of writing. Observe that

$$\begin{aligned}
X_{n+1}(H) &= \Delta_n S_{n+1}(H) + (1+r)(X_n - C_n - \Delta_n S_n) \\
&= \frac{V_{n+1}(H) - V_{n+1}(T)}{S_{n+1}(H) - S_{n+1}(T)}(S_{n+1}(H) - (1+r)S_n) + (1+r)(V_n - C_n) \\
&= \frac{V_{n+1}(H) - V_{n+1}(T)}{S_n(u-d)}(S_n)(u-1-r) + \tilde{p}V_{n+1}(H) + \tilde{q}V_{n+1}(T) \\
&= (V_{n+1}(H) - V_{n+1}(T))\tilde{q} + \tilde{p}V_{n+1}(H) + \tilde{q}V_{n+1}(T) \\
&= V_{n+1}(H)
\end{aligned}$$

as required. □

## §2.5. Markov Processes

Markov processes are a way to simplify calculations by determining what is relevant and what is not, akin to the computational "shortcuts" discussed in 1.4.

**Definition 2.5.1** (Markov Process)**.**
Consider the binomial asset-pricing model. Let $X_0, ..., X_N$ be an adapted process. If, for every $n \in \{0, ..., N-1\}$ and function $f(x)$, there is another function $g(x)$ (which we get given $n$ and $f$), such that

$$\mathbb{E}_n[f(X_{n+1})] = g(X_n)$$

we say that $X_0, ..., X_N$ is a Markov process.

In a bit, we prove that if such a $g$ exists, then the payoff of a derivative security is random only through its dependence on $X_N$, then we can create a pricing algorithm that does not need to store path information.

**Example 2.5.1.**
In the binomial model, where

$$S_{n+1}(\omega_1...\omega_{n+1}) = \begin{cases} uS_n(\omega_1...\omega_n) \text{ if } \omega_{n+1} = H \\ dS_n(\omega_1...\omega_n) \text{ if } \omega_{n+1} = T \end{cases}$$

we get that

$$\mathbb{E}_n[f(S_{n+1})](\omega_1...\omega_n) = pf(uS_n(\omega_1...\omega_n)) + qf(uS_n(\omega_1...\omega_n)).$$

The function $g(x) = pf(ux) + qf(dx)$ shows that this process is Markov.

To see why Markov processes are nice, consider the valuation of a derivative security with a payoff at time $N$ is $v_{N(S_N)}$ for some function $v_N$. Then, the price at time $n$ is given by

$$V_n = \frac{1}{1+r}\tilde{\mathbb{E}}_n[V_{n+1}], n = 0, ..., N-1.$$

But $V_N = v_N(S_N)$ and the stock price process is Markov, so

$$V_{N-1} = \frac{1}{1+r}\tilde{\mathbb{E}}_{N-1}[v_N(s_N)] = v_{N-1}(S_{N-1})$$

for some function $v_{N-1}$. We can continue generating such $v_n$ to get $V_n = v_n(S_n)$. Using the definition of $\tilde{\mathbb{E}}$, we get

$$v_n(s) = \frac{1}{1+r}[\tilde{p}v_{n+1}(us) + \tilde{q}v_{n+1}(ds)]$$

for $n = 0, ..., N-1$!

**Remark 2.5.1.**
Martingales are not necessarily Markov processes and Markov processes are not necessarily Martingales. For example, non-discounted stock prices are Markov processes but they are not Martingales, unless $pu + qd = 1$.

**Lemma 2.5.1** (Independence)**.**
Consider the $N$-period binomial asset pricing model with $n \in \{0, ..., N\}$. Suppose the random variables $X^1, ..., X^K$ depend only on coin tosses $1$ through $n$ and the random variables $Y^1, ..., Y^L$ depend only on coin tosses $n+1, ..., N$. Let $f(x^1, ..., x^K, y^1, ..., y^L)$ be a function of dummy variables $x^1, ..., x^K$ and $y^1, ..., y^L$, and define

$$g(x^1, ..., x^K) = \mathbb{E}f(x^1, ..., x^K, Y^1, ..., Y^L).$$

Then,

$$\mathbb{E}_n[f(X^1, ..., X^K, Y^1, ..., Y^L)] = g(X^1, ..., X^K).$$

If $K = L = 1$, this means if we set $g(x) = \mathbb{E}f(x, Y)$, then $\mathbb{E}_n[f(X, Y)] = g(X)$. This is the analog of "taking out what is known".

**Proof**: Let $\omega_1...\omega_n$ be fixed but arbitrary. By the definition of conditional expectation,

$$\mathbb{E}_n[f(X, Y)](\omega_1...\omega_n) = \sum_{\omega_{n+1}...\omega_N} f(X(\omega_1...\omega_n), Y(\omega_{n+1}...\omega_N))p^{\#H(\omega_{n+1}...\omega_N)}q^{\#T(\omega_{n+1}...\omega_N)}.$$

On the other hand,

$$g(x) = \mathbb{E}_n[f(x, Y)](\omega_1...\omega_n) = \sum_{\omega_{n+1}...\omega_N} f(x, Y(\omega_{n+1}...\omega_N))p^{\#H(\omega_{n+1}...\omega_N)}q^{\#T(\omega_{n+1}...\omega_N)}.$$

Which shows that

$$\mathbb{E}_n[f(X, Y)](\omega_1...\omega_n) = g(X(\omega_1...\omega_n)).$$

**Example 2.5.2.**
The maximum-to-date derivative is not a Markov process! $M_2(TH) = M_2(TT)$ but $M_3(TTH) = 4 \neq M_3(THH) = 8$ (which leads us to the discovery that $\mathbb{E}_3[M_3](TH))$ cannot equal $\mathbb{E}_3[M_3](TT))$.

Markov processes are much easier to compute. Is it possible to turn non-Markov processes Markov? The general idea is to add "state variables" to recover the Markov property.

**Definition 2.5.2** (Multi-dimensional Markov)**.**
Consider binomial asset-pricing model. Let $\{(X_n^1, ..., X_n^K); n = 0, ..., N\}$ be a $K$-dimensional adapted process; i.e., $K$ one-dimensional adapted processes. If, for every $n$ between $0$ and $N-1$ and for every function $f(x^1, ..., x^K)$, there is another function $g(x^1, ..., x^K)$ such that

$$\mathbb{E}_n[f(X_n^1, ..., X_n^K)] = g(X_n^1, ..., X_n^K),$$

we say that $\{(X_n^1, ..., X_n^K); n = 0, ..., N\}$ is a $K$-dimensional Markov process.

**Example 2.5.3.**
We can now turn the maximm-to-date derivative process Markov. Consider the two-dimensional

adapted process $\{(S_n, M_n); n = 0, ..., N\}$ where $S_n$ is the stock price and $M_n$ is the stock price maximum-to-date.

Let $Y = \frac{S_{n+1}}{S_n}$, only dependent on the $(n+1)$st coin flip. We want to compute

$$\mathbb{E}_n[f(S_{n+1}, M_{n+1})] = \mathbb{E}_n[f(S_n Y, \max(M_n, (S_n, Y)))].$$

If we let

$$g(s, m) = pf(us, \max(m, us)) + qf(ds, \max(m, ds))$$

We have found a $g$ so that $g(s, m) = \mathbb{E}[f(S_{n+1}, M_{n+1})]$.

**Remark 2.5.2.**
The Markov property is a "one-step-ahead" property. I.e. the Markov property implies that if $n \leq N - 2$, then for every function $h$ there exists a function $f$ so that

$$\mathbb{E}_{n+1}[h(X_{n+2})] = f(X_{n+1}).$$

This leads us to

$$\mathbb{E}_n[h(X_{n+2})] = \mathbb{E}_n[\mathbb{E}_{n+1}[h(X_{n+2})]] = \mathbb{E}_n[f(X_{n+1})]$$

or

$$\mathbb{E}_n[h(X_{n+2})] = g(X_n).$$

The upshot is that if $0 \leq n \leq m \leq N$ and $h(x^1, ..., x^K)$ is a function, there exists another function $g(x^1, ..., x^K)$ so that

$$\mathbb{E}_n[h(X_m^1, ..., X_m^K)] = g(X_m^1, ..., X_m^K)$$

Using the "one-step-ahead" property, we can simplify calculations for the valuation of derivatives. The idea is that we are guaranteed to have a function $v_n$ such that

$$\tilde{\mathbb{E}}_n\left[\frac{V_N}{(1+r)^{N-n}}\right](\omega_1 ... \omega_n) = v_n(X_n(\omega_1 ... \omega_n)) \quad \text{for all} \quad \omega_1 ... \omega_n,$$

meaning that

$$V_n = v_n(X_n)$$

i.e. there exists a function to calculate the value of the derivative. This is summarized in the following theorem:

**Theorem 2.5.1** (Markov Valuation).
Let $X_0, ..., X_N$ be a Markov process under risk-neutral probability measure $\tilde{\mathbb{P}}$ in the binomial model. Let $v_N(x)$ be a function and consider a security whose payoff at time $N$ is $v_N(X_N)$. Then, for each $n = 0, ..., N$, there exists a function $v_n$ of $X_n$ such that

$$V_n = v_n(X_n), n = 0, ..., N.$$

There is a recursive algorithm to compute such $v_n$.

**Example 2.5.4.**
Going back to the max-lookback stock option, we have

$$V_n = \frac{1}{1+r}\tilde{\mathbb{E}}_n[v_{n+1}(S_{n+1}, M_{n+1})] = \frac{1}{1+r}\tilde{\mathbb{E}}_n\left[v_{n+1}\left(S_n \cdot \frac{S_{n+1}}{S_n}, \max\left(M_n, S_n \cdot \frac{S_{n+1}}{S_n}\right)\right)\right].$$

Then,

$$v_n(s,m) = \frac{1}{1+r}\tilde{\mathbb{E}}_n\left[v_{n+1}\left(S_n \cdot \frac{S_{n+1}}{S_n}, \max\left(M_n, S_n \cdot \frac{S_{n+1}}{S_n}\right)\right)\right]$$

by the Markov property of the max-lookback stock option. We get

$$v_n(s,m) = \frac{1}{1+r}[\tilde{p}v_{n+1}(us, \max(m, us)) + \tilde{q}v_{n+1}(ds, m)]$$

Because $ds < m$ if $d \leq 1$ (usually assumed), giving us a recursive formula for $v_n$.

# State Prices
Volume 1, Chapter 3

## §3. State Prices

### §3.1. Change of Measure

So far, we have encountered the actual probabilities and the risk-neutral probabilities. Actual probabilities are useful because they reflect the real world. Risk-neutral probabilities are useful because under this measure, the discounted price of assets are martingales, helping us solve systems of equations.

**Definition 3.1.1** (Radon-Nikodym Derivative)**.**
Let $\Omega$ be a finite sample space with probability measures $\mathbb{P}$ and $\tilde{\mathbb{P}}$. Assume that in both $\mathbb{P}$ and $\tilde{\mathbb{P}}$, every element of $\Omega$ has positive probability. Define

$$Z(\omega) = \frac{\tilde{\mathbb{P}}(\omega)}{\mathbb{P}(\omega)}$$

Observe that $Z$ is a random variable! We call $Z$ the Radon-Nikodym derivative of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$.

**Theorem 3.1.1** (Properties of $Z$)**.**
1. $\mathbb{P}(Z > 0) = 1$
2. $\mathbb{E}Z = 1$
3. $\tilde{\mathbb{E}}Y = \mathbb{E}[ZY]$ for any random variable $Y$

**Proof**: 1. follows from assuming that $\tilde{\mathbb{P}}(\omega) > 0$ for all $\omega$. 2. comes from

$$\mathbb{E}Z = \sum_{\omega\in\Omega} Z(\omega)\mathbb{P}(\omega) = \sum_{\omega\in\Omega} \frac{\tilde{\mathbb{P}}(\omega)}{\mathbb{P}(\omega)}\mathbb{P}(\omega) = \sum_{\omega\in\Omega} \tilde{\mathbb{P}}(\omega) = 1.$$

3. comes from

$$\tilde{\mathbb{E}}Y = \sum_{\omega\in\Omega} Y(\omega)\tilde{\mathbb{P}}(\omega) = \sum_{\omega\in\Omega} Y(\omega)\frac{\tilde{\mathbb{P}}(\omega)}{\mathbb{P}(\omega)}\mathbb{P}(\omega) = \sum_{\omega\in\Omega} Y(\omega)Z(\omega)\mathbb{P}(\omega) = \mathbb{E}[ZY]. \qquad \square$$

The main power of the Radon-Nikodym derivative is that we can remove references to risk-neutral probabilities. Using $V_0 = \tilde{\mathbb{E}}\left[\frac{V_n}{(1+r)^n}\right]$ for valuations, we can instead write $V_0 = \mathbb{E}\left[\frac{V_n}{(1+r)^n}\right]$. This leads to the idea of state prices.

**Definition 3.1.2** (State Prices)**.**
In the $N$-period binomial model with actual probability measure $\mathbb{P}$ and risk-neutral probability measure $\tilde{\mathbb{P}}$, let $Z$ denote the Radon-Nikodym derivative of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$, i.e.

$$Z(\omega_1 \ldots \omega_n) = \frac{\tilde{\mathbb{P}}(\omega_1 \ldots \omega_n)}{\mathbb{P}(\omega_1 \ldots \omega_n)} = \left(\frac{\tilde{p}}{p}\right)^{\#H(\omega_1\ldots\omega_n)} \left(\frac{\tilde{q}}{q}\right)^{\#T(\omega_1\ldots\omega_n)}.$$

The state price density random variable is

$$\zeta(\omega) = \frac{Z(\omega)}{(1+r)^N}.$$

The **state price** corresponding to $\omega$ is $\zeta(\omega)\mathbb{P}(\omega)$.

Let $\overline{\omega}$ represent a specific sequence of coin flips $\overline{\omega}_1 ... \overline{\omega}_n$. Let $V_n(\omega) = 1$ if $\omega = \overline{\omega}$ and $0$ otherwise. Using the risk-neutral pricing formula, we get that the value of this derivative is

$$\tilde{\mathbb{E}}\frac{V_N}{(1+r)^N} = \frac{\tilde{\mathbb{P}}(\overline{\omega})}{(1+r)^N} = \frac{Z(\overline{\omega})\mathbb{P}(\overline{\omega})}{(1+r)^N} = \zeta(\overline{\omega})\mathbb{P}(\overline{\omega}).$$

The big idea from this equation is that if we did not discount for risk, then $\mathbb{P}(\overline{\omega})$ is the payoff at time $n$ and we have to discount the price and divide by $(1+r)^N$. But investors care about risk! This is where the $Z(\overline{\omega})$ term comes in handy: it accounts for risk.

To represent contracts that pay even with different events, we can just write

$$V_0 = \mathbb{E}[\zeta V_N] = \sum_{\omega \in \Omega} V_N(\omega)\zeta(\omega)\mathbb{P}(\omega)$$

## §3.2. Radon-Nikodym Derivative Process

Now, we want a way to estimate $Z$ based on information at time $n < N$. We first use the following theorem:

**Theorem 3.2.1.**
Let $Z$ be a random variable in an $N$-period binomial model. Define

$$Z_n = \mathbb{E}_n Z$$

for $n = 0, ..., N$. Then $Z_n$, $n = 0, ..., N$ is a martingale under $\mathbb{P}$.

**Proof**: Use iterated conditioning property:

$$\mathbb{E}_n[Z_{n+1}] = \mathbb{E}_n[\mathbb{E}_{n+1}Z] = \mathbb{E}_n[Z] = Z_n$$

as required. □

**Remark 3.2.1.**
This is true for the risk-neutral probability measure.

**Definition 3.2.1** (Radon-Nikodym Derivative Process)**.**
In the $N$-period binomial model with actual probability measure $\mathbb{P}$ and risk-neutral probability measure $\tilde{\mathbb{P}}$ with $\mathbb{P}(\omega), \tilde{\mathbb{P}}(\omega) > 0$ for all $\omega$. Let $Z$ denote the Radon-Nikodym derivative of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$. The **Radon-Nikodym** derivatve process is

$$Z_n = \mathbb{E}_n[Z], \quad n = 0, ..., N.$$

For the edge cases, $Z_N = Z$ and $Z_0 = 1$.

**Lemma 3.2.1.**
Let $Z_n$ be a Radon-Nikodym derivative process. Let $n$ be a positive integer between $0$ and $N$. Let $Y$ be a random variable depending only on the first $n$ coin tosses. Then

$$\tilde{\mathbb{E}}Y = \mathbb{E}[Z_n Y].$$

**Proof**: Use the theorems that $\tilde{\mathbb{E}}Y = \mathbb{E}[ZY]$, $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}_n[X]]$, and $\mathbb{E}[XY] = Y\mathbb{E}[X]$ for independent $Y$ in the following manipulation:

$$\tilde{\mathbb{E}}Y = \mathbb{E}[ZY] = \mathbb{E}\left[\mathbb{E}_{n[ZY]}\right] = \mathbb{E}[Y\mathbb{E}_n[Z]] = \mathbb{E}[YZ_n]. \qquad \square$$

**Example 3.2.1.**
Consider a derivative which pays $1$ at time $N$ if the first $n$ coin tosses are $\overline{\omega}_1 ... \overline{\omega}_n$ and $0$ otherwise. Then

$$\tilde{\mathbb{E}}Y = \tilde{\mathbb{P}}(\text{First } n \text{ coin tosses are } \overline{\omega}_1 ... \overline{\omega}_n) = \tilde{p}^{\#H(\overline{\omega}_1 ... \overline{\omega}_n)} \tilde{q}^{\#T(\overline{\omega}_1 ... \overline{\omega}_n)}$$

and

$$\begin{aligned}\mathbb{E}[YZ_n] &= Z_n(\overline{\omega}_1 ... \overline{\omega}_n)\mathbb{P}(\text{First } n \text{ coin tosses are } \overline{\omega}_1 ... \overline{\omega}_n) \\ &= Z_n(\overline{\omega}_1 ... \overline{\omega}_n) p^{\#H(\overline{\omega}_1 ... \overline{\omega}_n)} q^{\#T(\overline{\omega}_1 ... \overline{\omega}_n)}.\end{aligned}$$

By our lemma, these two quantities are equal:

$$\tilde{p}^{\#H(\overline{\omega}_1 ... \overline{\omega}_n)} \tilde{q}^{\#T(\overline{\omega}_1 ... \overline{\omega}_n)} = Z_n(\overline{\omega}_1 ... \overline{\omega}_n) p^{\#H(\overline{\omega}_1 ... \overline{\omega}_n)} q^{\#T(\overline{\omega}_1 ... \overline{\omega}_n)}$$

meaning that

$$Z_n(\overline{\omega}_1 ... \overline{\omega}_n) = \left(\frac{p}{\tilde{p}}\right)^{\#H(\overline{\omega}_1 ... \overline{\omega}_n)} \left(\frac{q}{\tilde{q}}\right)^{\#T(\overline{\omega}_1 ... \overline{\omega}_n)}$$

We have found a way to calculate $Z_n$!

We also want an analogue of our lemma for conditional expectations.

**Lemma 3.2.2.**
Let $Z_n$ be a Radon-Nikodym derivative process. Let $n \leq m$ be positive integers between $0$ and $N$. Let $Y$ be a random variable depending only on the first $n$ coin tosses. Then

$$\tilde{\mathbb{E}}_n Y = \frac{1}{Z_n}\mathbb{E}_n[Z_m Y].$$

**Proof**: Let $\omega_1 ... \omega_n$ be given. Compute:

$$\begin{aligned}
\tilde{\mathbb{E}}_n[Y](\omega_1...\omega_n) &= \sum_{\omega_{n+1}...\omega_m} Y(\omega_1...\omega_m)\tilde{p}^{\#H(\omega_{n+1}...\omega_m)}\tilde{q}^{\#T(\omega_{n+1}...\omega_m)} \\
&= \left(\frac{p}{\tilde{p}}\right)^{\#H(\omega_1...\omega_n)}\left(\frac{q}{\tilde{q}}\right)^{\#T(\omega_1...\omega_n)} \\
&\cdot \sum_{\omega_{n+1}...\omega_m}\left[Y(\omega_1...\omega_m)\left(\frac{p}{\tilde{p}}\right)^{\#H(\omega_1...\omega_n)}\left(\frac{q}{\tilde{q}}\right)^{\#T(\omega_1...\omega_n)}\right. \\
&\left. \cdot\, p^{\#H(\omega_{n+1}...\omega_m)}q^{\#T(\omega_{n+1}...\omega_m)}\right] \\
&= \frac{1}{Z(\omega_1...\omega_n)}\sum_{\omega_{n+1}...\omega_m} Y(\omega_1...\omega_m)Z_m(\omega_1...\omega_n)p^{\#H(\omega_{n+1}...\omega_m)}q^{(\#T(\omega_{n+1}...\omega_m)]} \\
&= \frac{1}{Z(\omega_1...\omega_n)}\mathbb{E}_n[YZ_m](\omega_1...\omega_n)\Bigg]
\end{aligned}$$

as required. □

We now have different ways of formulating the risk-neutral pricing formula.

**Theorem 3.2.2** (Different Formulations of Risk-Neutral Valuations)**.**
Consider $N$-period binomial model with $0 < d < 1 + r < u$. Let $p$ be the actual probability of a head and $q = 1 - p$ the actual probability of a tail. Then set:

$$\tilde{p} = \frac{1+r-d}{u-d}, \quad \tilde{q} = \frac{u-1-r}{u-d}$$

which are both positive. Let $\mathbb{P}$ and $\tilde{\mathbb{P}}$ be the actual and risk-neutral probability measures and let $Z$ be the Radon-Nikodym derivative of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$. Let $Z_n$, $n = 0, ..., N$ be the Radon-Nikodym derivative process. Let $\zeta_n$ be the state-price density process, i.e.

$$\zeta_n = \frac{Z_n}{(1+r)^n}, \quad n = 0, ..., N.$$

Consider a derivative security with payoff $V_N$ dependent on possibly all $N$ coin tosses. For $n = 0, ..., N$, the price at time $n$ of this derivative security is

$$V_n = \tilde{\mathbb{E}}_n \frac{V_N}{(1+r)^{N-n}} = \frac{(1+r)^n}{Z_n}\mathbb{E}_n \frac{Z_N V_N}{(1+r)^N} = \frac{1}{\zeta_n}\mathbb{E}_n[\zeta_N V_N].$$

**Proof**: The first equality is the risk-neutral pricing formula. The second is from our lemma. The third is from the definition of $\zeta_n$. □

## §3.3. Capital Asset Pricing Model

There are two ways of valuing assets. We have thoroughly explored the first method: no arbitrage. The second is the **capital asset pricing model** (CAPM), where we balance supply and demand. This method is not precise quantitatively but has many qualitative insights. CAPM is about balancing supply and demand in the economics sense, modelling investors as having utility functions that convert units of consumption to units of satisfaction.

**Definition 3.3.1** (Concavity).
Afunction $f$ is concave if its values lie above any secant line, i.e. for every $x, y \in \mathbb{R}$ and $\alpha \in (0,1)$,

$$f(\alpha x + (1-\alpha)y) \geq \alpha f(x) + (1-\alpha)f(y).$$

We say $f$ is strictly convex if the inequality is strict everywhere that it is finite.

**Definition 3.3.2** (Utility Function).
A **utility function** is a non-decreasing concave function defined on $\mathbb{R}$. It can take the value $-\infty$ but not $+\infty$.

**Example 3.3.1.**
$\ln(x)$ is a common example of a utlity function.

**Example 3.3.2.**
For $p < 1$, $p \neq 0$, and $c \in \mathbb{R}$, we can define a utility function

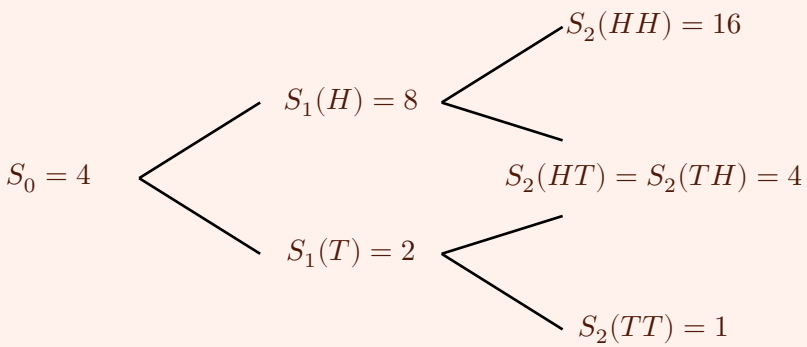
$$U_p = \begin{cases} \frac{1}{p}(x-c)^p & \text{if } x > c \\ 0 & \text{if } 0 < p < 1 \text{ and } x = c \\ -\infty & \text{if } p < 0 \text{ and } x = c \\ -\infty & \text{if } x < c \end{cases}$$

This is known as the HARA (hyperbolic absolute risk aversion) class.

**Remark 3.3.1.**
We require utility functions to be concave because it represents that there is a trade-off between risk and return (this happens as a result of Jensen's inequality, i.e. the expected value of the utility function will be less when the results are more "risky"). For example, a gamble with a $\frac{1}{2}$ probability of obtaining $\$1$ and a $\frac{1}{2}$ probability of obtaining $\$99$ has the same expected value as just getting $50$, but getting $50$ is more desirable because there is less risk.

The main idea in CAPM is the following problem: given initial wealth $X_0$ and a Utility function $U(X_N)$, find an adapted portfolio process $\Delta_0, ..., \Delta_{N-1}$ that maximizes $\mathbb{E}U(X_n)$, subject to $X_{n+1} = \Delta_n S_{n+1} + (1+r)(X_n - \Delta_n S_n)$ **(Problem CAPM 1)**.

**Example 3.3.3.**
Consider the following two-period model with $r = \frac{1}{4}$.




The risk-free rates are $\tilde{\mathbb{P}}(HH) = \tilde{\mathbb{P}}(HT) = \tilde{\mathbb{P}}(TH) = \tilde{\mathbb{P}}(TT) = \frac{1}{4}$, while the actual probability measures are $\tilde{\mathbb{P}}(HH) = \frac{4}{9}, \tilde{\mathbb{P}}(HT) = \tilde{\mathbb{P}}(TH) = \frac{2}{9}, \tilde{\mathbb{P}}(TT) = \frac{1}{9}$. Consider an agent that seeks to maximize $\mathbb{E}\ln(X_2)$, i.e. the agent has the utility function $\ln$.

After a few menial calculations, we arrive at

$$\begin{aligned}\mathbb{E}\ln X_2 &= \frac{4}{9}\ln(X_2(HH)) + \frac{2}{9}\ln(X_2(HT)) + \frac{2}{9}\ln(X_2(TH)) + \frac{1}{9}\ln(X_2(TT)) \\ &= \frac{4}{9}\ln\left(6\Delta_1(H) + \frac{15}{4}\Delta_0 + \frac{25}{4}\right) + \frac{2}{9}\ln\left(-6\Delta_1(H) + \frac{15}{4}\Delta_0 + \frac{25}{4}\right) \\ &+\frac{2}{9}\ln\left(\frac{3}{2}\Delta_1(T) - \frac{15}{4}\Delta_0 + \frac{25}{4}\right) + \frac{1}{9}\ln\left(-\frac{3}{2}\Delta_1(T) - \frac{15}{4}\Delta_0 + \frac{25}{4}\right).\end{aligned}$$

This expression expands out $\mathbb{E}U(X_2)$ with our adapted portfolio process. The arguments in each $\ln$ function represent $X_2$ down the various paths. To maximize this function w.r.t. our portfolio process (i.e. $\Delta_0$, $\Delta_1(T)$, and $\Delta_2(H)$), we write out $\frac{\partial}{\partial\Delta_0}\mathbb{E}\ln(X_2)$, $\frac{\partial}{\partial\Delta_1(T)}\mathbb{E}\ln(X_2)$, and $\frac{\partial}{\partial\Delta_2}\mathbb{E}\ln(X_2)$ and set them to zero to find the following equations:

$$\begin{aligned}\frac{4}{X_2(HH)} + \frac{2}{X_2(HT)} &= \frac{2}{X_2(TH)} + \frac{1}{X_2(TT)}, \\ \frac{2}{X_2(HH)} &= \frac{1}{X_2(HT)}, \\ \frac{2}{X_2(TH)} &= \frac{1}{X_2(TT)}.\end{aligned}$$

Through basic algebra, we arrive at

$$\begin{aligned}X_2(HH) &= 2X_2(HT), \\ X_2(TH) &= 2X_2(TT), \\ X_2(HT) &= 2X_2(TT)\end{aligned}$$

We then solve:

$$6\Delta_1(H) + \frac{15}{4}\Delta_0 + \frac{25}{4} = X_2(HH)$$
$$-6\Delta_1(H) + \frac{15}{4}\Delta_0 + \frac{25}{4} = X_2(HT)$$
$$\frac{3}{2}\Delta_1(T) - \frac{15}{4}\Delta_0 + \frac{25}{4} = X_2(TH)$$
$$-\frac{3}{2}\Delta_1(T) - \frac{15}{4}\Delta_0 + \frac{25}{4} = X_2(TT).$$

which yields $\Delta_0 = \frac{5}{9}, \Delta_1(H) = \frac{25}{54}$, and $\Delta_1(T) = \frac{25}{7}$.

We can also do this via $X_0 = \tilde{\mathbb{E}}\frac{X_N}{(1+r)^N}$ to quickly obtain the values for $X_2$ and then recursively obtain $\Delta_i$.

With the machinery we developed, we can solve a different problem to substitute our original one: given $X_0$, find a random variable that maximizes $\mathbb{E}U(X_N)$ subject to $\tilde{\mathbb{E}}\frac{X_N}{(1+r)^N} = X_0$ (**Problem CAPM 2**).

**Lemma 3.3.1.**
The problems are equivalent, i.e. suppose $\Delta_0^*, \Delta_1^*, ..., \Delta_{N-1}^*$ is an optimal portfolio process for CAPM problem 1 and $X_N^*$ is the optimal wealth random variable. Then, $X_N^*$ solves CAPM problem 2.

Conversely, if $X_N^*$ solves CAPM problem 2, the corresponding portfolio process $\Delta_0^*, \Delta_1^*, ..., \Delta_{N-1}^*$ that starts with $X_0$ and has value $X_N^*$ at time $N$ solves CAPM problem 1.

**Proof**: Assume that $\Delta_0^*, \Delta_1^*, ..., \Delta_{N-1}^*$ is an optimal portfolio process for CAPM problem 1 and $X_N^*$ is the corresponding optimal wealth random variable. Let $X_N$ be any random variable that satisfies the $\tilde{\mathbb{E}}\frac{X_N}{(1+r)^N} = X_0$ constraint. It suffices to show that $\mathbb{E}U(X_N) \leq \mathbb{E}U(X_N^*)$ ($X_N^*$ satsfies the CAPM problem 2 constraint because it is generated by a portfolio with initial wealth $X_0$). Observe that $X_N$ can be regarded as a **derivative security**. We have already shown how to replicate a derivative security with a portfolio process $\Delta_1, ..., \Delta_{N-1}$. This allows us compare the two processes: we can now use the fact that $X_N^*$ is the optimal wealth random variable to obtain $\mathbb{E}U(X_N^*) \geq \mathbb{E}U(X_N)$.

For the other direction, suppose $X_N^*$ satisfies CAPM problem 2. Construct the portfolio process $\Delta_0^*, \Delta_1^*, ..., \Delta_{N-1}^*$ to replicate it. Let $\Delta_0, ..., \Delta_{N-1}$ be another portfolio process that starts with initial wealth $X_0$ and leads to $X_N$ at time $N$. Because $X_N$ satisfies $\tilde{\mathbb{E}}\frac{X_N}{(1+r)^N} = X_0$ amd $X_N^*$ is the solution for CAPM probem 1, we conclude that $\mathbb{E}U(X_N^*) \geq \mathbb{E}U(X_N)$ and $X_N^*$ is indeed optimal in CAPM problem 1.

Throughout this chapter, we have also developed machinery to "hide" the risk-neutral easures with the Radon-Nikodym derivative. We can change the constraint of CAPM problem 2 to become

$$\mathbb{E}\frac{Z_N X_N}{(1+r)^N} = X_0 \iff \mathbb{E}\zeta_N X_N = X_0$$

where $\zeta_N = \frac{Z_N}{(1+r)^N}$. This gives rise to our final reformulation of the CAPM problem:

Let $\omega^1, ..., \omega^M$ be the possible **full** sequences of coin tosses and let $x_m = X_N(\omega_m)$. Observe here that $\omega^m$ is a sequence of coin tosses, not a singular one. Given $X_0$, we want to find the vector $(x_1, ..., x_M)$ that maximizes

$$\sum_{m=1}^{M} \mathbb{P}(\omega^m) U(x_m)$$

subject t

$$\sum_{m=1}^{M} \mathbb{P}(\omega^m) x_m \zeta_m = X_0$$

This formulation is quite powerful because it helps us find $X_N$ more easily. Namely, using Lagrangian multipliers, we have

$$\frac{\partial}{\partial x_m} \sum_{n=1}^{M} \mathbb{P}(\omega^n) U(x_n) = \lambda \mathbb{P}(\omega^m) \zeta_m x_m \iff U'(x_m) = \lambda \zeta_m$$

Because we assume $U$ is strictly concave and everywhere finite, its derivative is a decreasing function and thus has an inverse function $I$. We then get $x_m = I(\lambda \zeta_m)$, or $X_N(\omega^m) = I(\lambda \zeta_N(\omega^m))$. We can then put this into the constraint of our new problem:

$$\mathbb{E}\left[ \frac{Z}{(1+r)^N} I\left( \lambda \frac{Z}{(1+r)^N} \right) \right] = X_0$$

We can then solve for $\lambda$ to obtain $X_N$ and find the optimal portfolio process corresponding to $X_N$.

# American Options
Volume 1, Chapter 4

## §4. American Derivatives

### §4.1. Introduction

An **American option** allows for exercise at any time. This means that it has more (or at least equal) value than European options because there are more rights associated with the option. The problem that we run into with this instrument is that the process of an American option is a supermartingale under the risk-neutral measure.

**Definition 4.1.1** (Intrinsic Value)**.**
The intrinsic value of an option is the value from immediately exercising it.

To price an European option, we aim to replicate the option but we have to consider when it is best for the owner to exercise the option, or the **optimal exercise time**.

### §4.2. Non-Path-Dependent American Derivatives

The risk-neutral pricing formula developed in chapter 2 implies that the value of an American option is:

$$\begin{aligned} v_N(s) &= \max(g(s), 0) \\ v_{n(s)} &= \frac{1}{1+r}[\tilde{p}v_{n+1}(us) + \tilde{q}v_{n+1}(ds)], \quad n = N-1, N-2, ..., 0 \end{aligned}$$

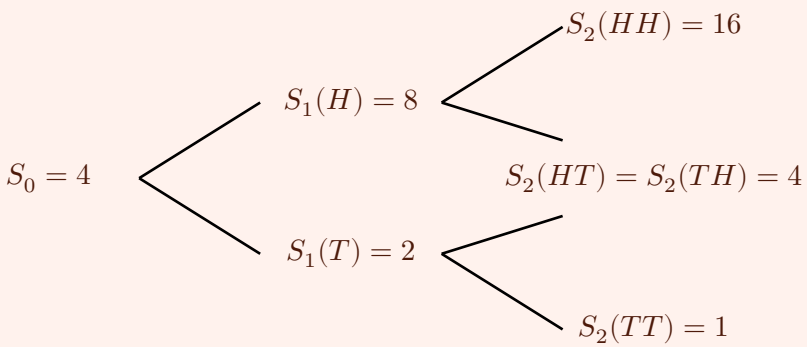
I.e. the risk-neutral expected value. The $\Delta$ for the replicating portfolio are defined by

$$\Delta_n = \frac{v_{n+1}(uS_n) - v_{n+1}(dS_n)}{(u-d)S_n}, \quad n = 0, ..., N.$$

Now, consider when the holder of an American option will exercise the option. They will exercise the option if the payoff, $g(S_n)$, from the option is higher than its valuation. This gives us the following modification:

$$\begin{aligned} v_N(s) &= \max(g(s), 0) \\ v_n(s) &= \max\left(g(s), \frac{1}{1+r}[\tilde{p}v_{n+1}(us) + \tilde{q}v_{n+1}(ds)]\right), \quad n = N-1, N-2, ..., 0 \end{aligned}$$

**Example 4.2.1.**
Consider the following tree:




To find the value of an American put option, expiring at time 2, with strike price 5, we have

$$v_2(16) = 0, v_2(4), \text{ and } v_2(1) = 4.$$

Now,

$$v_1(8) = \max\left(5 - 8, \frac{2}{5}(0 + 1)\right) = 0.40$$
$$v_1(2) = \max\left(5 - 2, \frac{2}{5}(1 + 4)\right) = 3$$

leading to

$$v_1(4) = \max\left(5 - 4, \frac{2}{5}(0.4 + 3)\right) = 1.36$$

If we price the corresponding European put option, we will see that the value of the American option is greater. This illustrates that American derivatives are going to form supermartingales under the risk-neutral probability measure.

To see how we can hedge optional exercise, suppose the first flip is a tail. If the owner chooses to exercise, then we are hedging. If the owner chooses to keep the option, the risk-neutral pricing formula states that we need a hedging portfolio valued at $\frac{2}{5}(v_2(4) + v_2(1)) = 2$, but our hedging portfolio is worth $3$. We can use $\$1$ however we like and still have $\$2$ to set up the hedge, where we sell $1$ share and invest $\$2.50$ in the money market.

**Theorem 4.2.1** (Replication of path-independent American derivatives)**.**
Consider an $N$-period binomial asset-pricing model with $0 < d < 1 + r < u$ and $\tilde{p} = \frac{1+r-d}{u-d}, \tilde{q} = \frac{u-1-r}{u-d}$. Let $g(s)$ be the payoff function. Define

$$v_N(s) = \max(g(s), 0)$$
$$v_n(s) = \max\left(g(s), \frac{1}{1+r}[\tilde{p}v_{n+1}(us) + \tilde{q}v_{n+1}(ds)]\right), \quad n = N-1, N-2, ..., 0$$

and

$$\Delta_n = \frac{v_{n+1}(uS_n) - v_{n+1}(dS_n)}{(u-d)S_n}$$
$$C_n = v_n(S_n) - \frac{1}{1+r}[\tilde{p}v_{n+1}(uS_n) + \tilde{q}v_{n+1}(dS_n)]$$

for $n = 0, ..., N-1$. Then, $C_n \geq 0$ for all $n$ and if $X_0 = v_0(S_0)$ and

$$X_{n+1} = \Delta_n S_{n+1} + (1+r)(X_n - C_n - \Delta_n S_n)$$

we have

$$X_n(\omega_1 ... \omega_n) = v_n(S_n(\omega_1 ... \omega_n))$$

for all $n$ and $\omega_1 ... \omega_n$. Further, $X_n \geq g(S_n)$ for all $n$.

This theorem is very similar to as in chapter 1 but adapted for American derivatives. $C_n \geq 0$ because $v_{n(s)} \geq \frac{1}{1+r}[\tilde{p} v_{n+1}(us) + \tilde{q} v_{n+1}(ds)]$ by definition. $X_n \geq g(S_n)$ because $X_{n(\omega_1 ... \omega_n)} = v_{n(S(\omega_1 ... \omega_n))} \geq g(s)$ by definition.

## §4.3. Stopping Times

We now define a new function to represent what time we should exercise given the sequence of events.

**Definition 4.3.1** (Stopping Time)**.**
Define the random variable $\tau : \Omega \to \{0, ..., N, \infty\}$, where $\tau(\omega_1 ... \omega_n)$ denotes when the owner should **exercise the option**. $\infty$ just means that the owner should let the option expire. Additionally, $\tau$ must satisfy the following condition:

- If $\tau(\omega_1 ... \omega_n \omega_{n+1} ... \omega_N) = n$, then $\tau(\omega_1 ... \omega_n \omega_{n+1}' ... \omega_N') = n$ for all $\omega_{n+1}' ... \omega_N'$.

**Example 4.3.1.**
If $N = 2$, $u = 2$, $d = \frac{1}{2}$, $r = \frac{1}{4}$, $S_0 = 4$, and we are valuing an American put option expiring at time $2$ with strike price $5$ (**our earlier example**), $\tau(TT) = \tau(TH) = 1$, $\tau(HT) = 2$, and $\tau(HH) = \infty$, i.e. if the first flip is a tails, we should exercise at time 1; if the first flip is a heads but the second flip is a tails, we should exercise at time 2; and if both flips are heads we should not exercise.

**Remark 4.3.1.**
The condition is given to ensure that the Stopping Time random variable does not depend on future information, i.e. once we stop, the variable does not rely on the future flips.

We now want a way to adapt our stochastic processes to "incorporate" early exercise and how the value of a process can be frozen after time $\tau$, which leads us to the stopped process.

**Definition 4.3.2** (Stopped Process)**.**
Let $a \wedge b = \min(a, b)$. Define the stopped process $Y_{n \wedge \tau}$ by

$$Y_{n \wedge \tau}(\omega_1 ... \omega_n) = \begin{cases} Y_n(\omega_1 ... \omega_n) & \text{if} \quad n \geq \tau(\omega_1 ... \omega_n) \\ Y_\tau(\omega_1 ... \omega_\tau) & \end{cases} \quad \text{if othersise.}$$

**Remark 4.3.2.**
We like stopped processes because the original process may be a supermartingale while the stopped process is a martingale. For example, the discounted American put price process $Y_n = \left(\frac{4}{5}\right)^n v_n(S_n)$ is a supermartingale because of the benefit of early exercise (in particular, $Y_1(T) = 2.40 \geq \frac{1}{2}Y_2(TH) + \frac{1}{2}Y_2(TT) = 1.60$), but the **stopped process is a martingale**:

$$2.40 = Y_{1\wedge\tau}(T) = \frac{1}{2}Y_{2\wedge\tau}(TH) + \frac{1}{2}Y_{2\wedge\tau}(TT) = \frac{1}{2}\cdot 2.40 + \frac{1}{2}\cdot 2.40.$$

We have the following theorems to formalize this remark.

**Theorem 4.3.1** (Optional Sampling Part 1)**.**
A martingale stopped at a stopping time is a martingale. A supermartingale [submartingale] stopped at a stopping time is also a supermartingale [submartingale].

**Proof**: Let $Y_1, ..., Y_N$ be a martingale. If $n < \tau$, then $Y_{n\wedge\tau} = Y_n = \mathbb{E}Y_{n+1} = \mathbb{E}Y_{n+1\wedge\tau}$. If $n \geq \tau$, then $Y_{n+1\wedge\tau} = Y_{n\wedge\tau}$, resulting in $Y_{n\wedge\tau} = \mathbb{E}Y_{n+1\wedge\tau}$. The proof for the supermartingale/submartingale process is similar.

**Remark 4.3.3.**
This theorem, which seems like quite a trivial result, does not hold true without the condition given in the definition of stopping times.

**Theorem 4.3.2** (Optional Sampling Part 2)**.**
Let $X_n$ for $n = 0, ..., N$ be a submartingale and let $\tau$ be a stopping time. Then $\mathbb{E}X_{n\wedge\tau} \leq \mathbb{E}X_n$. Likewise, if $X_n$ is a supermartingale, then $\mathbb{E}X_{n\wedge\tau} \geq \mathbb{E}X_n$. If $X_n$ is a martingale, then $\mathbb{E}X_{n\wedge\tau} = \mathbb{E}X_n$.

## §4.4. General Amercian Derivatives

We now consider American derivative securities that may be path-dependent.